7.3 Answers: fractions

← Unit 7.3 · Practice

If he gives an unsimplified answer that is correct (6/8 instead of 3/4), it's right. Say so first, then ask "Can it be simpler?"

Set 1 · Warm-up: equal fractions and "fraction of"

1. 1/2 (sharing between 2 gives bigger pieces than sharing between 4)
2. 2 (1/2 = 2/4)
3. 5 (1/2 = 5/10)
4. 6 (top and bottom × 2)
5. 6 (top and bottom × 3)
6. 4
7. 10
8. 15 (1/4 of 20 is 5, so 3/4 is 3 × 5)
9. 12 (1/5 of 30 is 6, so 2/5 is 12)
10. 3

Set 2 · Building: simplifying and mixed numbers

1. 1/2 (÷ 2)
2. 1/2 (÷ 5)
3. 3/4 (÷ 2)
4. 1/3 (÷ 4)
5. 2/3 (÷ 5)
6. 3/4 (÷ 4; or ÷ 2 twice, which is also fine)
7. 1 3/4 (4 quarters make 1 whole, 3 quarters left)
8. 3 2/3 (3 × 3 = 9, 2 thirds left)
9. 5/2 (2 wholes = 4 halves, plus 1 half)
10. 13/4 (3 wholes = 12 quarters, plus 1)
11. 1

Talk about it: when no number (except 1) divides both the top and the bottom.

Set 3 · Building: adding and subtracting

1. 3/5
2. 7/8 (the pieces stay eighths)
3. 2/3 (4/6, simplified)
4. 1 1/2 (6/4 = 1 2/4 = 1 1/2)
5. 3/4 (1/2 = 2/4)
6. 1/2 (1/3 = 2/6, so 2/6 + 1/6 = 3/6 = 1/2)
7. 1/4 (3/4 − 2/4)
8. 11/12 (8/12 + 3/12)
9. 5/6 (3/6 + 2/6)
10. 5/8 (1 = 8/8)
11. 1

Talk about it: 2/5 is *less* than 1/2, so adding something to 1/2 can't give 2/5. Halves and thirds are different sizes; cut both into sixths first.

Set 4 · Building: fraction × whole number, and mixed numbers

1. 1/2 (2/4)

2. 3/5
3. 2 2/3 (8/3: 6 thirds make 2 wholes, 2 thirds left)
4. 3 3/4 (15/4: 12 quarters make 3 wholes, 3 quarters left)
5. 4 (1 + 2 = 3 wholes, 1/2 + 1/2 = 1 more)
6. 4 1/4 (3 wholes; 3/4 + 2/4 = 5/4 = 1 1/4; 3 + 1 1/4)
7. 2 1/4 (3 − 1 = 2; 2/4 − 1/4 = 1/4)
8. 2 2/3 (4 − 1 = 3, then 3 − 1/3 = 2 2/3)
9. 2 1/4 litres (3 × 3/4 = 9/4)
10. a. 5/8 (1/4 = 2/8; 3/8 + 2/8) b. 3/8 (8/8 − 5/8)
11. 1

Set 5 · Stretch: fraction × fraction and word problems

1. 1/4 (half of a half)
2. 3/8 (half of 3/4: shade 3 of 4 strips, then half of them; 3 of 8 small pieces)
3. 1/2 (6/12 simplified; or “2/3 of 3 quarters is 2 quarters”)
4. 2/5 (6/15 simplified)
5. 1 1/4 (half of 2 is 1, half of 1/2 is 1/4)
6. 21 children (1/4 of 28 is 7, 3 × 7 = 21)
7. 1/2 litre (1 1/2 = 3/2; a third of 3 halves is 1 half)
8. 1/6 (a third of a half)
9. 3/4 hour = 45 minutes (2 1/4 − 1 1/2 = 2 1/4 − 1 2/4)
10. 25

Talk about it: “× 1/2” means “half of”. Taking part of something gives less than you started with.

Puzzle corner

- 3/4 = 1/2 + 1/4
- 5/6 = 1/2 + 1/3
- 1 = 1/2 + 1/3 + 1/6. With three *different* unit fractions this is the only way. (With four or more, there are lots: for example 1/2 + 1/4 + 1/6 + 1/12.)